

MATCHMINDAI - PROJECT PRESENTATION DECK

Real-Time Sports Intelligence & 10,000-Run Monte Carlo Match Prediction Engine

Live Web Application URL: matchmindai.streamlit.app	GitHub Source Repository: github.com/Soham-ss/MatchMindAI
Core Technology Stack: Python 3.14, Streamlit, Scikit-Learn, Plotly	Predictive Engine: 10,000 Monte Carlo Runs + ML RandomForest

10,000 Match Runs Simulated Per Ball	18 IPL Seasons Mapped 2008–2026	3 AI Sub-Agents Parallel Execution	100% Free Cloud Host Streamlit Community
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■ Slide 2: The Problems We Solved (Why We Built MatchMindAi)

Before MatchMindAi: Static tables, outdated historical records, raw URL dumps, no live weather or pitch awareness.

With MatchMindAi: 10,000-run ball-by-ball simulations, real-time Google News scrapers, and instant 1-line factual responses.

■ Slide 3: Our Solution (How MatchMindAi Fixes Everything)

■ **Why MatchMindAi Wins:** By blending **55% Monte Carlo stochastic math** with **45% Machine Learning pattern recognition** and live web scraping, MatchMindAi gives sports fans the highest level of predictive accuracy.

■ Slide 4: Main Platform Features & Highlights

■ Supported Chat Prompts:

- *'who wins 2018 ipl'* → ■ **IPL 2018 Winner:** Chennai Super Kings (CSK)
- *'who win orange cap in 2017'* → ■ **IPL 2017 Orange Cap:** David Warner (641 runs)
- *'In which team does Virat Kohli play?'* → ■ **Royal Challengers Bengaluru (RCB)**
- *'who won in 2005'* → ■■ **IPL History Note:** IPL was founded in 2008.

■ Slide 5: System Architecture & Workflow (How Data Flows)

■ Ensemble Win Probability Formula:

Final Win Probability = $(0.55 \times \text{Monte Carlo Win \%}) + (0.45 \times \text{RandomForest ML Win \%})$

■ Slide 6: Machine Learning & Simulation Math Explained Simply

■ Why Random Forest + Monte Carlo?

Machine learning alone can be rigid if a team has new players. Monte Carlo math simulates randomness (wickets, sixes, extras). Combining both ensures high accuracy even with squad changes.

■ Slide 7: Technical Challenges Faced & Solved

■■ **Key Takeaway:** Production deployment requires defensive programming — handling missing files, OS path differences, and non-standard user phrasing gracefully.

■ Slide 8: UI/UX & High-Energy Design System

- **User Delight:** Interactive CSS/JS particle animations make the application feel alive and responsive whenever the user moves their cursor.

■ Slide 9: Cloud Deployment & Infrastructure

- **Zero Hosting Cost:** Built using open-source tools and deployed on Streamlit Cloud's free tier, allowing unlimited public user access.

■ Slide 10: Conclusion & Future Roadmap

■ Conclusion	MatchMindAi successfully combines Artificial Intelligence, Machine Learning, Monte Carlo Simulations, and Real-Time Internet Scraping into a fast, accurate, and visually stunning web platform.
■ Live Score API Integration	Connect live cricket APIs for real-time win probability shifts during active live matches.
■ Fantasy Team Optimizer	Auto-generate optimal Dream11 / My11Circle lineups based on Monte Carlo expected points.
■ Multi-Sport Expansion	Extend simulation models to Football (Premier League, Champions League) and Basketball (NBA).